

SHANNON, Ireland

WMO: 039620

Lat: 52.700N

Lon: 8.917W

Elev: 46

StdP: 14.67

Time Zone: 1.00 (IRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.1	31.0	24.4	18.6	32.1	27.1	21.0	33.9	35.9	47.5	32.1	47.8	4.5	30	0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	74.2	63.6	70.9	62.1	68.3	60.9	65.5	71.3	63.7	68.4	62.3	66.5	8.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.1	86.7	67.9	61.8	82.6	66.1	60.4	78.7	64.7	30.2	71.1	29.0	68.5	28.0	66.6	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.7	25.0	22.1	DB	24.9	79.6	3.5	4.2	22.4	82.6	20.3	85.1	18.3	87.5	15.8	90.6	(4)
(5)			WB	24.2	68.0	3.5	2.3	21.7	69.6	19.6	71.0	17.7	72.3	15.2	74.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	43.3	43.7	45.7	49.3	53.7	58.3	61.0	60.7	57.7	52.8	47.2	43.9	(6)	
	DBStd	7.89	5.20	4.65	4.54	4.30	4.36	4.05	3.47	3.29	3.66	4.44	5.06	5.81	(7)	
	HDD50	938	212	179	144	62	12	0	0	0	1	23	109	196	(8)	
	HDD65	4955	674	595	599	471	350	206	132	141	222	379	533	653	(9)	
	CDD50	1485	4	4	11	41	128	249	342	331	231	110	26	8	(10)	
	CDD65	24	0	0	0	0	0	5	10	7	2	0	0	0	(11)	
	CDH74	130	0	0	0	0	7	47	49	26	1	0	0	0	(12)	
	CDH80	19	0	0	0	0	0	9	7	3	0	0	0	0	(13)	
(14) Wind	WSAvg	10.5	11.6	11.5	11.0	10.7	10.6	10.0	9.5	9.7	9.8	10.4	10.3	11.2	(14)	
(15) Precipitation	PrecAvg	38.1	4.0	3.0	2.8	2.4	2.5	2.6	2.6	3.3	3.2	3.6	4.0	4.2	(15)	
	PrecMax	50.1	8.7	8.1	5.3	4.4	5.6	6.6	5.3	9.1	6.6	6.9	10.4	9.2	(16)	
	PrecMin	27.9	0.4	0.0	0.0	0.4	0.4	0.4	0.5	0.2	0.4	0.0	1.7	0.0	(17)	
	PrecStd	5.1	1.9	1.8	1.2	1.1	1.2	1.3	1.1	1.7	1.4	1.5	1.6	1.9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.6	56.3	60.9	66.5	74.4	79.9	80.3	77.4	71.9	65.4	60.0	57.5	(19)	
		MCWB	52.8	52.5	52.6	56.2	62.1	65.5	65.7	67.4	63.6	60.2	56.8	54.9	(20)	
	2%	DB	53.5	54.0	56.5	61.9	68.9	73.0	73.7	71.7	68.6	62.5	57.7	55.3	(21)	
		MCWB	51.2	51.0	51.0	53.4	59.3	61.3	63.8	63.6	61.4	58.3	55.1	53.0	(22)	
	5%	DB	52.1	52.5	54.3	58.9	64.7	69.1	69.9	68.5	66.0	60.9	56.0	53.5	(23)	
		MCWB	50.1	49.9	49.9	52.3	56.3	59.9	62.4	61.8	59.9	57.4	53.7	51.3	(24)	
	10%	DB	50.8	51.0	52.7	56.5	61.3	65.7	67.3	66.4	64.0	59.3	54.4	51.9	(25)	
		MCWB	48.9	48.6	49.0	50.9	54.5	58.4	60.9	60.7	59.1	56.2	52.3	49.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.2	53.3	53.8	57.6	63.1	66.2	68.2	68.2	65.4	61.3	57.6	55.5	(27)	
		MCDB	54.8	55.2	57.5	64.1	72.4	76.5	77.0	76.0	70.1	63.7	59.2	57.0	(28)	
	2%	WB	51.7	51.6	52.2	54.9	60.1	62.9	65.4	65.3	63.0	59.4	55.6	53.3	(29)	
		MCDB	53.0	53.2	55.0	59.7	67.3	71.0	71.1	69.9	66.4	61.5	57.2	54.9	(30)	
	5%	WB	50.4	50.3	51.0	53.3	57.6	61.0	63.6	63.5	61.5	58.2	54.1	51.7	(31)	
		MCDB	51.8	52.2	53.7	57.5	62.7	67.1	68.1	67.0	64.6	60.3	55.6	53.1	(32)	
	10%	WB	49.1	48.9	49.6	51.7	55.7	59.2	62.0	62.0	59.9	56.7	52.7	50.1	(33)	
		MCDB	50.7	50.8	52.3	55.6	59.9	64.5	66.0	65.3	63.0	58.9	54.3	51.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	9.9	11.5	12.9	12.9	12.6	11.4	11.2	11.5	10.2	9.1	8.7	(35)	
		MCDBR	9.0	9.8	13.1	16.0	18.5	18.9	16.1	14.2	14.0	10.4	8.9	9.2	(36)	
	5% WB	MCWBR	8.5	8.4	9.0	9.7	10.3	9.8	8.7	8.2	8.5	7.8	7.9	8.6	(37)	
		MCWBR	9.0	9.3	11.5	13.8	16.3	16.3	14.0	12.3	11.7	9.7	8.7	9.2	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.341	0.349	0.371	0.380	0.393	0.377	0.382	0.364	0.360	0.346	0.334	0.331	(40)	
		taud	2.245	2.307	2.276	2.286	2.283	2.363	2.358	2.417	2.423	2.415	2.304	2.215	(41)	
	Ebn at Noon	Ebn at Noon	196	237	257	270	272	278	273	273	259	237	199	174	(42)	
		Edh at Noon	21	27	33	37	39	37	36	32	29	24	20	18	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	229	443	797	1243	1515	1601	1439	1226	903	531	277	172	(44)	
	RadStd	RadStd	24	37	75	108	90	137	116	85	65	45	23	22	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+148	N/A	N/A
(47) Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page